

Agentic AI for Information Retrieval

A Multi-Agent System for Multi-Hop Retrieval with
Self-Reflective Validation

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Abstract

Most retrieval-augmented systems retrieve once and generate once, and that structure cannot answer a question whose evidence is spread across documents, because no single document is rankable as the answer. This report presents an agentic alternative: four agents coordinated by an explicit state machine, covering all five tasks of the assignment brief. A Planner decomposes a question into a validated graph of dependent sub-queries, a Retrieval Agent routes each to BM25, dense or hybrid search, a Knowledge Graph Navigator recovers bridge entities from a graph induced from corpus text, and a Verifier scores the answer against its own citations. The central contribution is one backward edge: below its confidence threshold the Verifier returns a re-plan directive to the Planner instead of an answer. Everything runs locally on one consumer GPU with no paid API, and the edge is isolated by ablation rather than asserted.

The result is reported as measured. On answer accuracy the agentic system does not lead: exact match is 0.432 on HotpotQA and 0.224 on 2WikiMultihopQA against 0.504 and 0.264 for a self-ask baseline that decomposes but never verifies. The backward edge does measurable work on one dataset and none on the other. The largest effect in the ablation grid belongs to the component the brief lists first, and its sign is negative: deleting the Planner improves F1 by 0.121 on 2WikiMultihopQA at a third of the model calls. What the agentic stack earns its cost for is grounded, auditable answers, with 0.960 and 0.976 of citations resolving to real sentences where no baseline’s resolve at all.

A bounded improvement loop then tested five hypotheses against the one-node configuration. One survived: letting the synthesiser reason, with a completion budget large enough to hold it, takes exact match to 0.524 and 0.456, ties the strongest baseline on HotpotQA and beats it by 0.192 on 2WikiMultihopQA. The loop also established that seed pinning does not make this system reproducible, because a caching inference server changes the arithmetic between identical requests, and that finding bounds every comparison in this report.

Chapter 1

Introduction to Agentic AI in Information Retrieval

1.1 From Reactive Generation to Agency

A generative language model is a function from a prompt to a completion. It emits text and stops. It holds no goal that outlives the call, it cannot observe what its output caused, and it has no way of concluding that what it produced was not good enough. That describes a component, and it also describes most systems currently called retrieval-augmented: embed a query, fetch a ranked list, paste the list into a prompt, emit a completion. Retrieve once, generate once. Nothing in that structure allows the behaviour to be anything but reactive.

Agentic artificial intelligence names the class of systems built to remove that limitation. An agentic system pursues a goal across several steps, chooses its own actions, calls external instruments to carry them out, observes what came back, and revises its behaviour in light of the observation [14, 10]. Four properties carry the weight. **Autonomy**: the system, not the user, decides what happens next. **Goal-directed planning**: the work is represented explicitly before it is done, so the plan can be inspected, validated, repaired and discarded. **Tool integration**: the model is not the source of facts but the selector of instruments. **Adaptive reasoning**: the plan can be revised after execution has begun, because an observation can invalidate it.

The fourth property is the difficult one and the one most often missing. A system can be autonomous, planned and tool-using and still be a fixed pipeline whose stages happen to be model calls. The industry touchstones the assignment brief names, AutoGPT and Microsoft’s Jarvis, are notable for attempting the fourth property. The research literature shows the same direction. PaperQA retrieves across scientific articles, judges what it found and synthesises from the survivors [3]. Agentic Reasoning adds tool-using sub-agents and a mind-map agent [10]. ColLEX explores multimodal scientific collections [7], agentic contextual retrieval in telecommunications combines multi-source retrieval with self-reflective validation [13], and multi-agent geospatial copilots decompose remote-sensing workflows across specialised agents [4]. What they share is the shape, and what they leave open is the question this report exists to test: whether the feedback edge, once actually implemented, earns the compute it consumes.

1.2 The Limits of Static Ranking

Classical information retrieval is the study of a scoring function. Given a query and a corpus, rank the documents and return the top k . BM25 formalises this for lexical matching [6], dense bi-encoders replace term overlap with proximity in a learned space [11], reciprocal rank fusion combines rankings whose scores share no scale [1], and BEIR measures how well any of it transfers [8]. These are real and cumulative advances, and every one of them improves the ordering of a single list produced from a single query.

Consider a HotpotQA question [12] that this report reuses throughout:

When did the tour begin for the album Taylor Swift released on October 22, 2012?

The passage **Red (Taylor Swift album)** says the album was released on that date. The passage **The Red Tour** says the tour began on March 13, 2013. Of the 66,581 passages in the processed HotpotQA corpus, none contains both dates. That is not a statement about ranking quality. There is no document for a ranker to place first, because no single document answers the question. The bridge entity, the album title *Red*, does not occur in the question at all; it has to be derived before it can be searched for. A lexical retriever cannot match a term that was never typed, and a dense retriever embeds the whole question, so the sub-need “identify the album” is diluted by the sub-need “find its tour date” rather than issued on its own. This is the compositionality gap that Press et al. measure directly [5], and it is what motivates interleaving retrieval with reasoning rather than separating them [9].

Scale turns the difficulty into an impossibility. The two corpora hold 66,581 and 54,957 passages, so the number of ordered passage pairs is on the order of 10^9 . The composition cannot be materialised in advance; it has to be inferred from the question, which is a reasoning operation rather than a ranking one. Two pressures compound it: the distractor setting fills each candidate pool with passages chosen to look like the gold ones, and relevance for the first hop is defined only relative to an intermediate goal the user never stated.

1.3 Problem Statement

The problem is the design of a retrieval agent that, given a question and a static corpus, does four things a ranker does not. It interprets an intent whose surface form under-specifies the work. It decomposes that intent into ordered sub-goals with explicit dependencies, so that one retrieval’s result can be substituted into the next query. It orchestrates the resulting workflow, choosing an instrument for each sub-goal and traversing structure where the corpus admits one. And it judges its own answer against the evidence it actually retrieved before returning anything.

The fourth requirement is where a pipeline and an agent part company. The system described here is, with one exception, a forward pipeline: plan, execute, aggregate, synthesise. The exception is a single backward edge. A Verifier scores each candidate answer against the evidence cited for it, and when its confidence falls below a threshold it does not answer. It sends a **ReplanDirective** back to the Planner carrying what could not be established, which sub-queries returned nothing useful, and the text of every sub-query already tried, which the next plan may not repeat. One edge is a modest architectural change and a categorical behavioural one.

The system runs entirely on local hardware: **qwen3:8b** at Q4_K_M quantisation through Ollama on a single RTX 5060 with 8 GB of VRAM, measured at 52 tokens per second warm, with no API key anywhere. At that throughput a schema-constrained model call costs several seconds, so every call has to be justified against a deterministic alternative. Tool selection is a decision table over lexical features, entity linking is alias matching, graph traversal is breadth-first search, and the confidence blend is arithmetic. A system that cannot afford to think everywhere is forced to say where thinking pays, and, as Chapter 4 reports, the answer it arrived at was not the one it was designed around.

1.4 Objectives and Contributions

The objective is a working agentic retrieval system covering all five tasks of Table 1 of the assignment brief, whose central claim is tested rather than asserted. The contributions are four.

A four-agent architecture with full task coverage. A Planner emits a validated directed acyclic graph of sub-queries linked by placeholders. A Retrieval Agent selects among BM25, dense and hybrid search under a conditional reranking gate. A Knowledge Graph Navigator traverses an entity graph built from corpus text alone. A Verifier performs self-reflective validation. An orchestrator specified as an explicit state machine coordinates the four.

The Verifier-to-Planner feedback loop, isolated by ablation. A configuration with the backward edge disabled and everything else held fixed measures what the loop is worth, and the trace records which cycle produced the selected answer.

An evaluation protocol that prices agency. Answer and retrieval quality are reported beside model calls, tool calls, plan depth, re-plan rate and latency, so no gain appears without its cost. Nine configurations run over two multi-hop benchmarks, all eighteen cells on the same frozen 250 questions.

A local, reproducible implementation, and an honest account of how reproducible. Every model runs on-device, prompts are versioned files whose rendered hash is logged, and a test asserts that no module reaches the network. Section 4.8.4 reports the measured limit of that reproducibility, which turned out to be a finding in its own right.

Table 1.1 records where each task of the brief is implemented and measured. One clause of the brief is deliberately not met: it mentions orchestrating external APIs for real-time enrichment, and this system exposes no external API at all, because a tool whose output changes between runs makes the evaluation unrepeatable and the trace unfalsifiable. The registry is written so that such a connector could be added as one more entry.

Table 1.1: The five tasks of the brief’s Table 1, the component implementing each, and where the report specifies and measures it. Four agents cover five tasks because query refinement and retrieval planning are one act performed twice: before any evidence exists, and again after the Verifier has said what is missing.

Task (brief, Table 1)	Agent or component	Specified in; measured in
Autonomous query refinement	Planner: per-node rewrites; re-plan on a directive	§3.3.1, §3.3.4; re-plan rate in Table 4.5
Multi-step retrieval planning	Planner: validated sub-query DAG; orchestrator state machine	§3.3.1, §4.1; §4.5
Tool-augmented retrieval	Retrieval Agent: registry, routing table, reranking gate	§3.3.2; Table 4.5
Knowledge graph traversal	KG Navigator: alias linking, bidirectional BFS	§3.3.3, §2.5; §4.5
Self-reflective validation and fact checking	Verifier: NLI entailment, citation grounding, feedback edge	§3.3.4; §4.5, §4.6

A fifth outcome was not designed but is reported because it is useful. During development the 8B model produced correctly shaped decompositions in 8 of 8 cases with no format retries, and mislabelled its own plan’s **strategy** field in 7 of those 8. The system therefore derives the strategy from the shape of the validated graph and discards what the model claimed. Where a small model’s self-report can be trusted, and where it cannot, is a practical result for anyone building agents at this scale.

Chapter 2 covers data and preprocessing, Chapter 3 the retrieval methodologies and why single-shot retrieval fails on compound needs, Chapter 4 the implementation, evaluation, ablations, error analysis and the improvement loop that followed, and Chapter 5 what the evidence supports and what follows.

Chapter 2

Data and Preprocessing

2.1 Requirements for Evaluating an Agentic IR System

A dataset is chosen because it can answer the questions a report has to ask. This system’s central claim, that a validation agent can send useful feedback to a planning agent, is meaningful only if it can be falsified with numbers. Four requirements follow, stated before any dataset is named.

Queries with relevance judgements. Recall@ k , nDCG@10 and MRR are defined over a mapping from queries to relevant documents. Without one those metrics are undefined, and the comparison against baseline IR systems that Chapter 4 must present cannot be computed.

Genuinely compositional needs. On single-fact look-ups the correct decomposition is the identity, every plan has one node, and the agentic apparatus collapses onto the hybrid baseline it is supposed to beat. Measuring a planner needs questions no single document answers.

Entity and structure annotations. Sentence-level supporting facts make retrieved evidence checkable at the granularity the system emits; gold relational triples make a graph traversal itself checkable.

A corpus that indexes and runs on consumer hardware. Every component runs on a single 8 GB laptop GPU with no paid API. A web-scale corpus is not expensive under that constraint; it is unusable.

BEIR [8] satisfies the first and fourth requirements across eighteen tasks, but most of its tasks express single-hop needs with no reasoning-structure annotations. The two datasets adopted below satisfy all four, and the first of them is itself one of BEIR’s tasks.

2.2 On the Datasets Listed in the Assignment Brief

Table 2 of the brief lists The Pile, Common Crawl, WikiText, OpenWebText and LAION-5B. All five are pretraining corpora: unlabelled text, or weakly labelled image and text pairs, at a scale suitable for fitting a model. None ships a query set, a relevance judgement or a gold answer, so the first requirement fails outright. This is not a deficiency in those corpora but a statement of what they are for. They are, in fact, the right resources for a different layer of this system. The 8B model that does the reasoning here owes its grasp of English, of world entities and of the shape of a JSON object to pretraining on exactly such corpora. The brief’s list describes the substrate the agent brings to the task; it does not describe the benchmark retrieval is scored against. The brief’s own instruction to check for updated datasets is the licence to select evaluation data on evaluation criteria.

2.3 HotpotQA

HotpotQA [12] is a multi-hop question answering dataset over English Wikipedia in which each question requires facts from two distinct paragraphs. This work uses the **distractor** setting, where each question carries its two gold paragraphs and eight distractors, and the **validation** split of 7,405 questions.

Its distinguishing property is the supporting-facts annotation: every question carries the (`title`, `sent_id`) pairs a human annotator used. This is retrieval ground truth at sentence granularity, and it fixes the granularity of the rest of the system. Evidence is carried as sentences, so that what the Verifier reasons over and what the metrics score are the same objects.

Reading the contexts of all 7,405 questions yields 73,700 paragraphs, which reduce to 66,581 distinct passages after deduplication; 7,119 discarded paragraphs were byte-identical repeats and no title carries two body texts. The corpus holds 269,602 sentences, a mean of 4.049 per passage. Of 18,005 gold supporting facts, 18,004 resolve; the exception cites sentence 902 of a five-sentence paragraph, an upstream annotation error reported rather than dropped, because the same counter would catch a real identifier drift.

One measured property governs the sampling design. HotpotQA labels questions easy, medium or hard, but every one of the 7,405 validation questions is **hard**; the easier levels exist only in training. The informative partition is reasoning type: 5,918 bridge questions, where one entity must be resolved before the second hop can be posed, and 1,487 comparison questions, where two entities are retrieved and an attribute compared.

2.4 2WikiMultihopQA

2WikiMultihopQA [2] was built to expose the reasoning steps a question requires, not only its endpoints. Its validation split holds 12,576 questions whose 125,760 context paragraphs reduce to 54,957 distinct passages and 202,759 sentences. It ships 30,687 sentence-level supporting facts, all of which resolve, and, in addition, 31,120 gold Wikidata evidence triples covering every question. Those triples convert the KG Navigator from a component this report describes into one it measures.

They also introduce the most serious leakage risk in the project. A graph built from those triples would let the KG Navigator read the answer key. The graph is therefore built from passage text only, the triples are loaded exclusively by scoring code, and a test asserts that the graph builder never opens a file whose name contains **evidence** or **triple**.

All four reasoning types are populated: compositional (5,236), comparison (3,040), bridge-comparison (2,751) and inference (1,549). The third is the hardest case in either dataset, requiring a bridge resolution and a comparison over its results. The split is also textually messier: because the same article appears in many contexts with inconsistent detokenisation, 1,720 titles ship two or more variants, producing 2,617 title collisions, and since `sent_id` indexes a sentence list the variant kept decides whether a gold fact lands on the right sentence.

2.5 Preprocessing Pipeline

The pipeline is deliberately thin. It applies only transformations that cannot be undone downstream and declines every one a consumer can perform for itself.

Segmentation and identifiers. Both datasets ship each paragraph as a sentence list, and that list is adopted verbatim. No sentence splitter is called anywhere, because the list is the identifier space of the ground truth: re-splitting would renumber every gold fact while the metrics went on producing plausible numbers. Each passage is keyed by `source:quote(title)`, so a retrieved passage, a graph node and a gold fact name the same object the same way.

Deduplication. HotpotQA collapses cleanly. 2Wiki leaves 2,617 cases where one title carries two texts; the rule keeps the variant with the most sentences, the widest identifier space, and its cost is measured: keeping the first occurrence would leave 70 gold identifiers out of range and 211 on the wrong sentence, whereas keeping the widest leaves none out of range and 150, or 0.489%, on a slightly different detokenisation of the intended sentence.

What is deliberately not done. The only text transformation is the collapse of whitespace runs. There is no lowercasing, stopword removal, stemming or punctuation stripping at the corpus level, because those are properties of a scorer’s analyser: BM25 stems inside its own analyser, the dense encoder needs original case to distinguish proper nouns, and the NLI model needs well-formed sentences. One convention is shared by all scorers: every one sees the same title-prefixed unit, "Title. body", because fusing a channel that indexes titles with one that does not produces rankings that cannot be compared.

Entity graph and indexes. Nodes are normalised passage titles whose aliases are compiled into a trie for longest-match linking. A directed edge $A \rightarrow B$ is created when an alias of B occurs in the text of A , and the edge stores the sentence asserting it, which is what makes a traversal result citable at the same granularity as a retrieved passage. The HotpotQA graph retains 64,733 nodes and 123,759 edges; the 2Wiki graph 52,959 nodes and 63,073 edges. A document-frequency cap removed 446 and 172 hub nodes, titles such as **one** whose occurrences are not relational assertions. Each corpus is also indexed twice, by BM25 with Snowball stemming and by an exact FAISS index over `bge-small-en-v1.5` embeddings; the two occupy 129.1MB and 103.6MB, small enough to hold in memory, which is why exact search is used and no recall is lost at the index level.

2.6 Evaluation Sampling

Evaluation runs on a frozen 250-question sample per dataset, and the reason is arithmetic. At 52 tokens per second a full agentic question was estimated before the runs at 22 to 25 seconds; the medians actually measured came out higher, and the estimate is recorded as the estimate it was. On it, one agentic configuration over the full 7,405-question split is about 45 hours, and the grid is nine configurations over two datasets. At 250 questions the grid was feasible on this hardware; at 7,405 it was not.

The sample is drawn once with seed 42, materialised to disk, and never re-sampled; its SHA-256 is recorded in every run’s metadata so a silently changed sample is detectable afterwards. Stratification is on question type, not difficulty, and that is a measured decision: with every HotpotQA validation question labelled hard, stratifying on level would give a single stratum, an unstratified sample presented as a stratified one. Stratifying on type preserves the bridge and comparison ratio exactly at 200 and 50 questions, and 2Wiki’s four types at 104, 60, 55 and 31. Table 2.2 gives the composition.

Two hundred and fifty questions is enough to separate the systems compared here. Treated as an unpaired proportion near 0.5, the 95% half-width is about 6.2 points. The comparisons that matter are paired: every configuration answers the identical 250 questions, so the bootstrap is taken over per-question differences, and the correlation between systems sharing a corpus, an index and a question set makes those intervals materially tighter. Each dataset also gets a disjoint 50-question calibration slice, verified to share no question identifier with the 250, reserved for setting the Verifier’s threshold. Table 2.1 summarises the two corpora as built.

Table 2.1: Corpus statistics after preprocessing, read from `{dataset}_corpus_stats.json`. Passages are de-duplicated by content under a canonical title-derived id; sentence splits are the datasets’ own and are persisted verbatim, because gold supporting facts index into them by position and re-splitting anywhere downstream would renumber the ground truth while every metric still looked plausible. - marks a statistic the build did not record.

Statistic	hotpotqa	twowiki
Questions in split	7 405	12 576
Paragraphs read	73 700	125 760
Passages after de-duplication	66 581	54 957
Identical paragraphs dropped	7 119	68 186
Title collisions resolved	0	2 617
Sentences	269 602	202 759
Mean sentences per passage	4.05	3.69
Mean passage length (chars)	531.9	427.2
Mean sentence length (chars)	130.6	115.1
Questions with gold triples	0	12 576
Gold supporting facts	18 005	30 687
Supporting facts unresolvable	1	0
Evaluation sample	250	250

Table 2.2: Composition of the frozen evaluation slice against the full split. The slice is stratified on question type and sampled once, with seed 42, by `scripts/sample_eval_set.py`; its SHA-256 is recorded in every run’s `meta.json` and it is never re-sampled at run time. HotpotQA’s validation split is entirely `level=hard`, so difficulty gives one stratum and a silently unstratified sample; question type is the informative split there.

Question type	Split <i>n</i>	Split share	Sample <i>n</i>	Sample share
<i>hotpotqa</i>				
bridge	5 918	0.799	200	0.800
comparison	1 487	0.201	50	0.200
<i>total</i>	7 405	1.000	250	1.000
<i>twowiki</i>				
bridge_comparison	2 751	0.219	55	0.220
comparison	3 040	0.242	60	0.240
compositional	5 236	0.416	104	0.416
inference	1 549	0.123	31	0.124
<i>total</i>	12 576	1.000	250	1.000

Chapter 3

Retrieval Methodologies

3.1 Classical Foundations

The agentic components do not replace classical retrieval; they schedule it. Every sub-query the Planner emits is answered by one of the three mechanisms below, so their blind spots determine what the agent layer has to compensate for.

3.1.1 Sparse Retrieval: BM25

BM25 [6] scores a document against a query by summing, over query terms, a saturating function of term frequency weighted by inverse document frequency:

$$\text{score}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}} \right)}.$$

The values used here, $k_1 = 0.9$ and $b = 0.4$, sit below the classical 1.2 and 0.75 on purpose: the indexed unit is a single lead paragraph averaging four sentences, where repetition carries little extra evidence and length differences are incidental. BM25 matches rare surface forms exactly, needs no training and answers in milliseconds. Its weakness is vocabulary mismatch: a query and a relevant passage sharing no stemmed term score zero against each other however obviously they concern the same thing.

3.1.2 Dense Retrieval

A dense retriever embeds queries and passages independently and ranks by similarity. This work uses `bge-small-en-v1.5` [11], a 384-dimensional bi-encoder with L2-normalised vectors in an exact FAISS index, used asymmetrically as its authors intend. Its strength is paraphrase, exactly what BM25 cannot see; its weakness is the mirror image. Query and passage never meet, and each is compressed to 384 dimensions before comparison, a bottleneck that preserves topic far better than identity, so two similarly named people are liable to be conflated. The two failure modes are close to complementary, which is the standing argument for combining them.

3.1.3 Hybrid Fusion and Reranking

The two lists are combined by reciprocal rank fusion [1], which discards scores and combines positions:

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}, \quad k = 60.$$

Rank fusion is preferred over score normalisation because BM25 scores are unbounded and corpus-dependent while cosine similarities occupy a narrow band; the two share no scale, and a manufactured one makes the fused score depend on the shape of the candidate list. Fusion improves recall and leaves head precision to chance, which is what reranking addresses. A cross-encoder,

`ms-marco-MiniLM-L-6-v2`, runs full attention over each query and candidate pair; nothing about the passage can be precomputed, so scoring n candidates costs n forward passes. That fixes the depths at 50 before reranking and 10 after. When the fused top result already leads the runner-up by a relative margin of 0.15 the fifty passes cannot change the head and are skipped, a saving counted in the trace rather than assumed.

3.2 Why Single-Shot Retrieval Fails on Multi-Hop Needs

Everything above computes a scoring function $s(q, d)$. The three methods differ enormously in expressive power but not in their arguments: all assume that the relevance of a document is determinable from the query and that document. For a bridge question the assumption is false, and the consequence is not that these systems perform poorly but that they solve the wrong problem.

Consider question `5a7793275542992a6e59df04` from the evaluation sample: *Which country is the firm that owns Babycham located?* The gold answer is Australia, and the two gold sentences open the passages **Babycham** (“...the name is now owned by Accolade Wines”) and **Accolade Wines** (“...headquarters in South Australia”). The first hop flatters BM25: Babycham is a near-unique term in a 66,581-passage corpus, and lexical matching alone puts the passage at rank one. The second hop is not merely harder. Of the query terms, not one occurs in the second gold sentence. BM25 has nothing to match. A bi-encoder produces a non-zero similarity, but the query embedding is dominated by Babycham and a possessive relation while the passage says wine business, global, South Australia, and nothing distinguishes it from the corpus’s other company profiles.

The reranker does not close the gap, for a reason that survives any pool width and any model size. Place the Accolade Wines passage in the pool by fiat and give the cross-encoder unlimited capacity. The information that would justify a high score, that Accolade Wines owns Babycham, is present in neither element of the pair it is shown; it is asserted in a third document. The relevance of the second-hop passage is a conditional quantity $\text{rel}(q, d_2 \mid d_1)$ depending on a document the ranker was never given, and more capacity changes the function class, not the arguments. The query that retrieves Accolade Wines trivially is “Where is Accolade Wines located?”, and it cannot be written in advance because that string exists only as the output of the first retrieval. The missing capability is a control structure, not a scoring function.

This is not an edge case. Bridge questions are 5,918 of HotpotQA’s 7,405, and 2Wiki’s compositional, inference and bridge-comparison types, 9,536 of 12,576 questions, share the conditional structure. Comparison questions fail differently: both entities are named in the question, so both passages are individually retrievable, and the difficulty is coverage at a fixed cut-off plus the fact that the answer is an operation over two passages rather than the content of either. Section 3.3.1 gives the two shapes different plan topologies accordingly.

3.3 Agentic Retrieval

Figure 3.1 shows the system this section builds up to. The remedy is a retrieval process that carries state between rounds, which is the thread through the agentic IR literature: a model that plans, invokes tools and revises [14, 10], whether over scientific literature [3], multimodal collections [7], telecommunications networks [13] or geospatial workflows [4]. The four agents below cover the five tasks of the brief’s Table 1.

3.3.1 Query Decomposition and Refinement

The Planner converts a question into a directed acyclic graph of at most five sub-queries. Each node carries its text, a dependency list, an intent and an expected answer type, and may contain placeholders such as `{{q1.answer}}`. Execution proceeds in topological order, and a node’s placeholders are substituted from its dependencies immediately before it is routed, so the second-hop query is a genuinely different string at retrieval time from the one the Planner emitted. This is the territory of Self-Ask [5] and IRCoT [9], with four differences. The plan is a validated data structure rather than a decoding

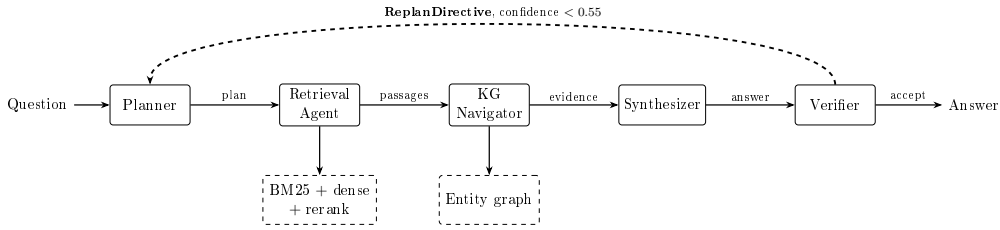


Figure 3.1: The four-agent architecture. Solid arrows are the forward retrieval path; dashed boxes are offline artefacts. The dashed backward edge is the Verifier-to-Planner feedback loop, the only information that flows against the pipeline.

trajectory, and every mechanical repair is recorded. It is a graph, so a comparison question yields two independent roots retrieved together. Query expansion rides inside the same planner response, so it costs no additional call. And the plan can be revised on evidence from a downstream agent, which no single-pass trajectory can do.

Two decisions followed from measurement. The plan’s **strategy** label is derived from the shape of the validated graph and overwrites what the model reported: on eight representative questions the model produced structurally correct decompositions 8 times out of 8 and labelled the strategy **single_hop** in 7 of them. It builds the graph well and names it badly. Separately, the model reliably emits a final combiner node restating the question and depending on every other node; detecting and skipping it removes a third of all retrieval on a three-node comparison plan.

The Planner degrades along a deterministic ladder rather than failing: a comparison template, then a two-node bridge template, then an identity plan routed to hybrid search with reranking. That floor is not the **hybrid_rerank** baseline, and the ablation grid of Chapter 4 shows why the distinction matters: an identity plan still passes through synthesis and verification, so the system with the Planner removed is a one-node agentic system that answers and cites, not a ranker.

3.3.2 Tool-Augmented Retrieval

Seven tools are declared in a registry, four over the indexes and three over the entity graph, and the description that documents each to a human is the string injected into the selection prompt, so documentation and registry cannot drift apart.

Selection among the retrieval tools is a seven-rule decision table over features of the resolved sub-query: token count, quoted spans, capitalised runs, out-of-vocabulary rate against the BM25 lexicon, minimum document frequency among content terms, and whether the node has dependencies. A planner hint is honoured first (R1); an out-of-vocabulary rate above 0.34 routes to dense search (R2); short quoted or proper-noun-dense queries (R3) and queries containing a term of document frequency three or fewer (R4) route to BM25, the Babycham case; dependent nodes (R5) and long or entity-free queries (R6) route to hybrid; and a default row (R7) makes the table total.

One fact about the table changes what the argument is about. In the headline HotpotQA run the Planner attached a hint to every one of the 854 sub-queries it emitted, so R1 fired on all 575 routing decisions and R2 through R7 never executed. All seven rules are implemented and tested; on this data six are dead code. The planning call the system pays for anyway already carries a per-node tool choice, and the table’s contribution was to avoid asking the model twice. The saving is real, but the claim that lexical rules decide as a model would have decided is untested. Reinforcement learning would enter here and does not, because the only reward derives from gold supporting facts that exist solely in the evaluation data.

3.3.3 Knowledge Graph Traversal

The KG Navigator operates on the entity graph of Section 2.5. Linking is an alias-trie longest match and costs no model call. Traversal is bounded at two hops and twenty-five neighbours, expanded in deterministic order of descending edge weight. The highest-value operation is bidirectional breadth-first search: given two or more linked seeds it expands from each and reports the meeting node as the

bridge entity. Applied to the Babycham example, the alias Accolade Wines occurring in the Babycham passage is an edge by construction, so the bridge entity the retrieval stack cannot surface by ranking is recovered by a graph walk in milliseconds. This is the concrete sense in which traversal is not decorative: it is a second, structurally different mechanism for the one operation single-shot retrieval provably cannot perform.

Traversal is also the most explainable component, because the path is the explanation: every edge stores the sentence asserting it, so a traversal result is scored by the same metrics as a retrieved passage. The limits are equally clear. The graph is induced from mention co-occurrence, and 23.2% of HotpotQA edges and 21.0% of 2Wiki edges carry no label beyond **mentions**, so it is a navigation aid rather than a knowledge base.

3.3.4 Self-Reflective Validation

The Verifier receives the Synthesizer’s answer with the evidence pool and decides whether it is supported. Cited identifiers that do not exist are recorded as hallucinated citations and contribute no support. A natural language inference model, **cross-encoder/nli-deberta-v3-base**, is run with each cited sentence as premise and the answer sentence as hypothesis; the Synthesizer emits that sentence as a separate field precisely so the hypothesis is a well-formed declarative sentence. Two further signals are computed, citation grounding and retrieval agreement, and the four combine as

$$\text{conf} = 0.45 \text{ nli} + 0.25 \text{ citation} + 0.15 \text{ retrieval} + 0.15 \text{ llm},$$

renormalised when the model term is absent. That term is gated: the adjudication call is issued only when the confidence computed without it falls inside $[0.40, 0.70]$, since confident acceptances and rejections cannot have their verdict changed by it.

Above the threshold of 0.55 the verdict is accept. Below it, with re-plans remaining, the Verifier emits a directive carrying what could not be established, which sub-queries returned nothing useful, and the normalised text of every sub-query already tried. Without that ban list an 8B model told only that its previous attempt was insufficient reliably re-emits the same decomposition and burns both re-plans for nothing. Below the threshold with no re-plans left the verdict is abstain, a label rather than a refusal: the best candidate is still returned. Answer selection at termination is the argument maximising confidence over all cycles, ties broken toward the earlier answer, so a bad re-plan cannot silently overwrite a good first attempt, and the error taxonomy of Chapter 4 contains explicit false-accept and false-reject categories so the loop is falsifiable.

3.4 The Role of the Language Model

The language model is the component that generalises, and everything else exists to keep the number of calls small. Three decisions genuinely require it: mapping an unseen compound question onto a dependency structure, composing a short answer from heterogeneous evidence, and adjudicating the narrow band where the numeric signals disagree. Everything else is deterministic. The worst case over a question is 15 calls against a cap of 20, with four to seven typical, and two calls are reserved so synthesis and verification always run.

Serving the model locally shaped the method, not only the deployment. Prompt length is a latency cost, which is why the re-plan prompt passes retrieved titles rather than text. And **qwen3** is a hybrid reasoning model that emits a **<think>** block by default: with thinking enabled and a 128-token cap it spent the entire budget on a 503-character reasoning block and produced no answer at all. At that budget thinking starves the response, so it is suppressed for the evaluated grid, and Section 4.8 reports what happens when it is switched back on for one agent with a budget large enough to hold it.

3.5 Architectural Summary

Read left to right, Figure 3.1 is a pipeline; the dashed edge, along which the Verifier returns a directive to the Planner below a confidence of 0.55, is what makes it an agent. Every other component is standard machinery arranged carefully, and Chapter 4 isolates that edge’s effect by ablation.

Chapter 4

System Implementation and Evaluation

Chapter 3 argued that a multi-hop need cannot be served by a single retrieval round and set out the four agents that answer it. This chapter describes how they are built, how the system is measured, and what the measurements show. It closes with a bounded improvement loop, because a negative result is more useful when the obvious remedies have been tried.

4.1 System Architecture

The orchestrator is an explicit state machine rather than a chain of function calls. Nine states, from `INIT` through `PLAN`, `EXECUTE`, `SYNTHESIZE` and `VERIFY` to `DONE`, and fifteen labelled transitions are enumerated in the source, and the transition sequence each question took is written into its trace. This costs more code than implicit control flow and buys inspectability: the claim that the system re-plans is checkable against a recorded path, and every early exit is attributable to a named guard.

Only one transition runs backwards. `REPLAN_GATE` returns to `PLAN` when the Verifier’s confidence is below threshold. An agent that can send work backwards can loop, so five guards are evaluated in order, on the re-plan count, the iteration count, the remaining calls, the elapsed time, and whether the directive names anything missing. A sixth check rejects a revised plan whose sub-queries pair with the previous plan’s at a Jaccard similarity of 0.85 or above, because a local 8B model told its attempt was insufficient will frequently re-emit the same decomposition in different words.

4.2 Implementation Details

All generation runs on `qwen3:8b` at `Q4_K_M` quantisation through Ollama on an RTX 5060 laptop GPU with 8 GB of VRAM. Measured throughput is 52 tokens per second warm, and a schema-constrained planning call averages 3.75 s. The card is effectively full: the model and its 8192-token context occupy 6707 of 8151 MiB, so the three sentence-transformer encoders run on CPU at query time, where they cost well under a second per question. The configuration file declares a response cache; none is implemented, so every latency reported here is an uncached measurement.

Every agent calls the model with a JSON schema and parses the result through a five-rung ladder ending in a re-prompt. Each question carries a budget of 20 calls, 6 iterations and 300 s, checked and never raised as an exception, so that running out of calls is distinguishable from crashing. Two calls are reserved for synthesis and verification, so a planner that decomposes aggressively cannot leave nothing with which to answer.

All results use the frozen 250-question samples of Section 2.6. Every run writes a JSON Lines trace with one record per question, a metadata file capturing the full configuration, library versions, model digest and seed, and a flat metrics file. The harness checkpoints by appending, so a run interrupted at question 194 resumes at 194. Sorting is deterministic with an explicit tiebreaker on document identifier. The implementation is covered by 444 automated tests, three of which enforce properties that would otherwise fail silently: no module outside the model client can reach the network, no agent reads the gold answer, and the graph builder never opens a gold evidence file.

4.3 Baselines and Metrics

Five baselines run from weakest to strongest. **bm25_only** and **dense_only** isolate the two channels. **hybrid_rerank** fuses them and reranks the top 50 with the cross-encoder on every question, with the margin gate switched off so the baseline is not weakened. **naive_rag** retrieves once with the same pipeline and makes one generation call, so the difference between it and **hybrid_rerank** is one generation step and nothing else. **self_ask** is the published multi-hop method [5]: decomposition with no verification and no re-planning, which makes the gap between it and the full system the closest available estimate of what the backward edge is worth.

Three families of metric are reported together. Retrieval quality is Recall@{2, 5, 10}, nDCG@10 and MRR, computed with **pytrec_eval**, which wraps the same core as the evaluation that produced the published BEIR figures [8]. Answer quality is exact match and token F1 under the official HotpotQA normalisation [12], with supporting-fact EM and F1 over sentence identifiers. Agent-specific cost is model calls, calls avoided by deterministic rules, tool calls, latency, plan depth, re-plan rate and citation grounding. Reporting cost beside quality is what the brief asks for and the only honest way to present an agentic system: a method that gains three points of F1 for twenty times the compute has not obviously won. Differences between systems are assessed with a paired bootstrap over 1000 resamples of the identical question set.

4.4 Results

Table 4.1: Answer quality on **hotpotqa**, n questions each. EM and F1 follow the official HotpotQA script; SP columns are set metrics over supporting-fact sentences. Reference system for the EM and F1 columns: **self_ask**, marked †. **Bold** = significantly better than the reference, † = significantly worse (paired bootstrap, 95% CI of the paired difference excluding zero). Unmarked differences are not significant. SP-EM, SP-P, SP-R and SP-F1 are not one protocol: **agentic_full**, **agentic_no_planner**, **agentic_no_kg** and **agentic_no_verifier** are scored on the sentences cited (mean 1.5 per question) and **bm25_only**, **dense_only**, **hybrid_rerank**, **naive_rag** and **self_ask** on the whole evidence pool (mean 6.9), so no significance mark is placed between rows of different protocol; *Pool SP-R* is scored the same way for every row. - across a whole row marks a configuration that has not been run. **bm25_only**, **dense_only** and **hybrid_rerank** are retrieval-only: they emit a ranking and no answer, so the EM, F1 and $\Delta F1$ columns show - because there is no answer to score, not because the run is absent. Scoring the unattempted answer would report 0.000 and assert a failed attempt that never happened. * marks **naive_rag** and **self_ask**, whose citations do not resolve, so their supporting-fact scores are computed over the whole evidence pool rather than a cited subset.

System	n	EM	F1	F1 95% CI	SP-EM	SP-P	SP-R	SP-F1	Pool SP-R	$\Delta F1$ vs. ref. [95% CI]
<i>Baselines (weakest first)</i>										
bm25_only	250	-	-	-	0.012	0.245 [↓]	0.588 [↓]	0.330 [↓]	0.588 [↓]	-
dense_only	250	-	-	-	0.020	0.283 [↓]	0.650 [↓]	0.377 [↓]	0.650 [↓]	-
hybrid_rerank	250	-	-	-	0.016	0.308 [†]	0.706 [†]	0.412 [†]	0.706 [†]	-
naive_rag	250	0.496	0.581	[0.522, 0.641]	0.016*	0.308*	0.706*	0.412*	0.706	-0.022 [-0.060, +0.018]
self_ask	250	0.504 [†]	0.603 [†]	[0.547, 0.659]	0.016*	0.302*	0.696*	0.405*	0.696	-
<i>Agentic system and ablations</i>										
agentic_full	250	0.432 [↓]	0.510 [↓]	[0.451, 0.571]	0.088	0.654	0.406	0.481	0.793	-0.093 [-0.145, -0.039]
agentic_no_planner	250	0.456	0.543 [↓]	[0.489, 0.602]	0.136	0.685	0.431	0.508	0.817	-0.060 [-0.106, -0.009]
agentic_no_kg	250	0.404 [↓]	0.501 [↓]	[0.443, 0.565]	0.088	0.635	0.386	0.460	0.783	-0.102 [-0.153, -0.051]
agentic_no_verifier	250	0.396 [↓]	0.474 [↓]	[0.418, 0.536]	0.108	0.672	0.414	0.491	0.795	-0.130 [-0.185, -0.074]

Tables 4.1 and 4.3 report answer and supporting-fact quality, Tables 4.2 and 4.4 retrieval quality, and Tables 4.5 and 4.6 the cost columns. Every figure is generated from the run artefacts and inserted verbatim; no number in this report is typed by hand from a run. A configuration that was not run renders as a dash rather than as zero, and a retrieval-only system renders a dash in the answer columns, because a zero there would assert a failed attempt that never happened.

The retrieval ladder behaves as Chapter 3 predicts. On HotpotQA the sparse channel reaches Recall@10 of 0.774, the dense channel 0.816, and fusing them with reranking 0.876, six points over the better single channel without any model involvement. Answer quality does not follow that ladder, and the first thing to record is that the agentic system does not win. On HotpotQA **self_ask** reaches exact match 0.504 and F1 0.603, **naive_rag** 0.496 and 0.581, and **agentic_full** 0.432 and 0.510. On

Table 4.2: Retrieval quality on **hotpotqa**, computed over the reciprocal-rank fusion of every sub-query’s ranking against the gold supporting-fact documents. Reference system for the retrieval columns: **hybrid_rerank**, marked †. **Bold** = significantly better than the reference, † = significantly worse (paired bootstrap, 95% CI of the paired difference excluding zero). Unmarked differences are not significant. - across a whole row marks a configuration that has not been run.

System	R@2	R@5	R@10	nDCG@10	nDCG@10 95% CI	MRR
<i>Baselines (weakest first)</i>						
bm25_only	0.576	0.700	0.774 [↓]	0.728 [↓]	[0.696, 0.754]	0.873
dense_only	0.652	0.762	0.816 [↓]	0.787 [↓]	[0.756, 0.817]	0.917
hybrid_rerank	0.696	0.824	0.876 [†]	0.835 [†]	[0.810, 0.859]	0.935
naive_rag	0.696	0.824	0.876	0.835	[0.810, 0.859]	0.935
self_ask	0.688	0.838	0.888	0.837	[0.812, 0.860]	0.929
<i>Agentic system and ablations</i>						
agentic_full	0.602	0.742	0.812 [↓]	0.763 [↓]	[0.734, 0.791]	0.899
agentic_no_planner	0.686	0.820	0.872	0.831 [↓]	[0.804, 0.855]	0.933
agentic_no_kg	0.604	0.746	0.820 [↓]	0.766 [↓]	[0.737, 0.793]	0.897
agentic_no_verifier	0.612	0.756	0.796 [↓]	0.758 [↓]	[0.729, 0.786]	0.888

Table 4.3: Answer quality on **twowiki**, n questions each. EM and F1 follow the official HotpotQA script; SP columns are set metrics over supporting-fact sentences. Reference system for the EM and F1 columns: **self_ask**, marked †. **Bold** = significantly better than the reference, † = significantly worse (paired bootstrap, 95% CI of the paired difference excluding zero). Unmarked differences are not significant. SP-EM, SP-P, SP-R and SP-F1 are not one protocol: **agentic_full**, **agentic_no_planner**, **agentic_no_kg** and **agentic_no_verifier** are scored on the sentences cited (mean 1.5 per question) and **bm25_only**, **dense_only**, **hybrid_rerank**, **naive_rag** and **self_ask** on the whole evidence pool (mean 8.9), so no significance mark is placed between rows of different protocol; *Pool SP-R* is scored the same way for every row. - across a whole row marks a configuration that has not been run. **bm25_only**, **dense_only** and **hybrid_rerank** are retrieval-only: they emit a ranking and no answer, so the EM, F1 and $\Delta F1$ columns show - because there is no answer to score, not because the run is absent. Scoring the unattempted answer would report 0.000 and assert a failed attempt that never happened. * marks **naive_rag** and **self_ask**, whose citations do not resolve, so their supporting-fact scores are computed over the whole evidence pool rather than a cited subset.

System	n	EM	F1	F1 95% CI	SP-EM	SP-P	SP-R	SP-F1	Pool SP-R	$\Delta F1$ vs. ref. [95% CI]
<i>Baselines (weakest first)</i>										
bm25_only	250	-	-	-	0.008	0.208 [↓]	0.551 [↓]	0.273 [↓]	0.551 [↓]	-
dense_only	250	-	-	-	0.020	0.244	0.605	0.314	0.605	-
hybrid_rerank	250	-	-	-	0.016	0.255 [†]	0.622 [†]	0.327 [†]	0.622 [†]	-
naive_rag	250	0.152 [↓]	0.200 [↓]	[0.155, 0.246]	0.016*	0.255*	0.622*	0.327*	0.622	-0.110 [-0.160, -0.063]
self_ask	250	0.264 [†]	0.310 [†]	[0.259, 0.364]	0.016*	0.246*	0.627*	0.318*	0.627	-
<i>Agentic system and ablations</i>										
agentic_full	250	0.224	0.267	[0.216, 0.321]	0.124	0.558	0.367	0.421	0.684	-0.043 [-0.100, +0.014]
agentic_no_planner	250	0.324	0.388	[0.329, 0.443]	0.144	0.678	0.422	0.495	0.678	+0.078 [+0.021, +0.135]
agentic_no_kg	250	0.256	0.285	[0.229, 0.339]	0.132	0.578	0.363	0.425	0.679	-0.025 [-0.079, +0.033]
agentic_no_verifier	250	0.216	0.266	[0.218, 0.319]	0.148	0.612	0.403	0.465	0.689	-0.044 [-0.102, +0.014]

Table 4.4: Retrieval quality on **twowiki**, computed over the reciprocal-rank fusion of every sub-query’s ranking against the gold supporting-fact documents. Reference system for the retrieval columns: **hybrid_rerank**, marked †. **Bold** = significantly better than the reference, † = significantly worse (paired bootstrap, 95% CI of the paired difference excluding zero). Unmarked differences are not significant. - across a whole row marks a configuration that has not been run.

System	R@2	R@5	R@10	nDCG@10	nDCG@10 95% CI	MRR
<i>Baselines (weakest first)</i>						
bm25_only	0.551	0.641	0.693 [↓]	0.698 [↓]	[0.671, 0.724]	0.910
dense_only	0.605	0.687	0.714	0.751	[0.729, 0.774]	0.981
hybrid_rerank	0.622	0.696	0.727 [†]	0.756 [†]	[0.730, 0.780]	0.967
naive_rag	0.622	0.696	0.727	0.756	[0.730, 0.780]	0.967
self_ask	0.627	0.756	0.797	0.790	[0.765, 0.817]	0.934
<i>Agentic system and ablations</i>						
agentic_full	0.559	0.683	0.727	0.723 [↓]	[0.697, 0.748]	0.917
agentic_no_planner	0.621	0.698	0.731	0.757	[0.732, 0.781]	0.967
agentic_no_kg	0.571	0.685	0.736	0.732 [↓]	[0.707, 0.757]	0.925
agentic_no_verifier	0.566	0.684	0.719	0.721 [↓]	[0.696, 0.744]	0.915

Table 4.5: Agent-specific cost on **hotpotqa**, per question. *Saved* counts calls a deterministic rule replaced where a model path was configured. *Cite grounding* is averaged over answered questions. Latency is the median per question with the maximum beside it, because one blocked call can dominate a mean. **One or more questions overran the 300 s orchestrator.max_wall_clock_s budget: agentic_full** (65744.3 s on 5a7af32e55429931da12c99c, 18.3 h), **agentic_no_kg** (36447.4 s on 5ae603ba5542996de7b71af4, 10.1 h); the budget is tested between calls and cannot pre-empt one that blocks. *Saved* is recomputed from the step trace: a call counts only where a configured model path would have issued it and a rule pre-empted it. Per-question terms, counted ones in bold: **agentic_full**: routing 3.31, KG alias 2.54, **verifier gate 1.21**, verifier pointless 0.04, **extraction 1.39** (agents recorded 7.10); **agentic_no_planner**: routing 1.00, KG alias 0.94, **verifier gate 0.64**, verifier pointless 0.01, planner ablation 1.19 (agents recorded 3.78); **agentic_no_kg**: routing 3.28, **verifier gate 1.11**, verifier pointless 0.04, **extraction 1.40** (agents recorded 4.42); **agentic_no_verifier**: routing 2.26, KG alias 1.59, **extraction 1.16** (agents recorded 3.85). *Plan depth* is that of the selected cycle, which differs from the latest plan’s wherever a re-plan executed and the first answer was kept (**agentic_full** 1.71 selected against 1.56 latest, **agentic_no_kg** 1.75 selected against 1.60 latest). *Re-plan rate* is the fraction of questions on which a re-plan was triggered (*trig.*) and on which one executed (*exec.*); a near-duplicate re-plan is discarded before it runs. Questions with a discarded re-plan: **agentic_full** (4 of 250 questions, executed rate 0.472), **agentic_no_planner** (47 of 250 questions, executed rate 0.000), **agentic_no_kg** (4 of 250 questions, executed rate 0.444). - across a whole row marks a configuration that has not been run.

System	LLM calls	Saved	Tool calls	Latency (s) med.	Latency (s) max	Plan depth	Sub-queries	Re-plan rate (trig.)	Re-plan rate (exec.)	Cite grounding
<i>Baselines (weakest first)</i>										
bm25_only	0.00	0.00	1.00	0.0	0.0	1.00	1.00	0.000	0.000	0.000
dense_only	0.00	0.00	1.00	0.1	16.4	1.00	1.00	0.000	0.000	0.000
hybrid_rerank	0.00	0.00	2.00	4.9	36.8	1.00	1.00	0.000	0.000	0.000
naive_rag	1.00	0.00	2.00	8.7	31.5	1.00	1.00	0.000	0.000	0.000
self_ask	1.24	0.00	2.41	4.5	35.7	1.20	1.20	0.000	0.000	0.000
<i>Agentic system and ablations</i>										
agentic_full	4.26	2.60	14.83	28.3	65744.3	1.71	1.80	0.472	0.472	0.960
agentic_no_planner	1.35	0.64	3.94	9.7	37.4	1.00	1.00	0.188	0.000	0.936
agentic_no_kg	4.28	2.51	8.15	11.8	36447.4	1.75	1.84	0.444	0.444	0.968
agentic_no_verifier	2.06	1.16	10.88	16.5	46.9	1.98	2.34	0.000	0.000	0.000

Table 4.6: Agent-specific cost on **twowiki**, per question. *Saved* counts calls a deterministic rule replaced where a model path was configured. *Cite grounding* is averaged over answered questions. Latency is the median per question with the maximum beside it, because one blocked call can dominate a mean. *Saved* is recomputed from the step trace: a call counts only where a configured model path would have issued it and a rule pre-empted it. Per-question terms, counted ones in bold: **agentic_full**: routing 3.68, KG alias 2.32, **verifier gate 1.16**, verifier pointless 0.03, **extraction 1.94** (agents recorded 7.19); **agentic_no_planner**: routing 1.00, KG alias 0.87, **verifier gate 0.70**, verifier pointless 0.04, planner ablation 1.09 (agents recorded 3.70); **agentic_no_kg**: routing 3.70, **verifier gate 1.14**, verifier pointless 0.04, **extraction 1.93** (agents recorded 4.88); **agentic_no_verifier**: routing 2.72, KG alias 1.54, **extraction 1.62** (agents recorded 4.25). *Plan depth* is that of the selected cycle, which differs from the latest plan’s wherever a re-plan executed and the first answer was kept (**agentic_full** 1.90 selected against 1.76 latest, **agentic_no_kg** 1.86 selected against 1.75 latest). *Re-plan rate* is the fraction of questions on which a re-plan was triggered (*trig.*) and on which one executed (*exec.*); a near-duplicate re-plan is discarded before it runs. Questions with a discarded re-plan: **agentic_full** (3 of 250 questions, executed rate 0.392), **agentic_no_planner** (22 of 250 questions, executed rate 0.000), **agentic_no_kg** (6 of 250 questions, executed rate 0.412). - across a whole row marks a configuration that has not been run.

System	LLM calls	Saved	Tool calls	Latency (s) med.	Latency (s) max	Plan depth	Sub-queries	Re-plan rate (trig.)	Re-plan rate (exec.)	Cite grounding
<i>Baselines (weakest first)</i>										
bm25_only	0.00	0.00	1.00	0.0	0.0	1.00	1.00	0.000	0.000	0.000
dense_only	0.00	0.00	1.00	0.1	13.9	1.00	1.00	0.000	0.000	0.000
hybrid_rerank	0.00	0.00	2.00	8.3	27.0	1.00	1.00	0.000	0.000	0.000
naive_rag	1.00	0.00	2.00	12.2	29.7	1.00	1.00	0.000	0.000	0.000
self_ask	1.46	0.00	2.86	13.2	40.3	1.43	1.43	0.000	0.000	0.000
<i>Agentic system and ablations</i>										
agentic_full	3.80	3.10	16.00	35.4	121.5	1.90	2.26	0.392	0.392	0.976
agentic_no_planner	1.26	0.70	3.88	4.5	30.9	1.00	1.00	0.088	0.000	0.900
agentic_no_kg	3.87	3.07	8.65	15.8	64.6	1.86	2.25	0.412	0.412	0.956
agentic_no_verifier	2.03	1.62	12.34	18.4	74.7	2.10	2.73	0.000	0.000	0.000

2WikiMultihopQA the leader is unchanged, **self_ask** at 0.264 and 0.310 against **agentic_full** at 0.224 and 0.267. Referenced to **self_ask**, the paired difference in F1 is -0.093 with a 95% interval of $[-0.145, -0.039]$ on HotpotQA, a deficit that is established, and -0.043 $[-0.100, +0.014]$ on 2WikiMultihopQA, a consistent point estimate that is not. On the answer string, decomposition with verification loses to decomposition without it.

Retrieval shows the same deficit from the other side. **agentic_full** records nDCG@10 of 0.763 on HotpotQA against 0.835 for **hybrid_rerank**, and 0.723 on 2WikiMultihopQA against 0.756, both significantly worse. The reported ranking for a multi-hop configuration is the fusion of every sub-query’s list, and narrow sub-queries each relevant to one hop dilute the head of the combined list. On 2WikiMultihopQA the two systems tie exactly on Recall@10 at 0.727 while the agentic nDCG@10 is three points lower: the same documents are found, further down, spread across several lists.

Supporting-fact F1 is the metric on which the trade is favourable, and it has to be read with care. **agentic_full** reaches 0.481 on HotpotQA against 0.412 for **hybrid_rerank**, and 0.421 against 0.327 on 2WikiMultihopQA, but the agentic rows are scored on the sentences the system cited, a mean of 1.5 per question, while the baselines are scored on their whole evidence pool, a mean of 6.9 and 8.9. The two protocols trade precision against recall in opposite directions and one F1 across them does not rank the systems; the tables place no significance mark between rows of different protocol. The one supporting-fact column scored the same way for every row is the recall of the whole pool, and on it the agentic system leads every baseline: 0.793 against 0.706 on HotpotQA and 0.684 against 0.622 on 2WikiMultihopQA, both significant. What is categorical is citation grounding: 0.960 and 0.976 for **agentic_full** and 0.000 for every baseline, whose citations resolve to nothing. The defensible claim from this grid is about grounded and auditable answers rather than accuracy: the system returns a somewhat worse answer with a precise, machine-checkable account of itself.

The cost columns price that account. **agentic_full** spends 4.26 model calls per question on HotpotQA and 3.80 on 2WikiMultihopQA against 1.24 and 1.46 for **self_ask**, and issues 14.83 and 16.00 tool calls against 2.41 and 2.86. Deterministic rules displace a real minority of the inference a fully model-driven version would spend, 2.60 and 3.10 calls per question; they do not displace more than the three reasoning agents consume. Latency is printed as a median with the maximum beside it. On 2WikiMultihopQA the full system costs a median of 35.4s against 13.2s for **self_ask**; on HotpotQA 28.3s against 4.5s, and the maximum is 65744s, 18.3 hours on one question inside a single Synthesizer call. No arrangement of four model calls consumes that; the host was almost certainly suspended with the call open. The measurement identifies a defect in the budget itself: the 300s guard is tested between nodes and before calls, never during one, so it cannot pre-empt a call that blocks.

4.5 Ablation Study

Table 4.7: Ablation study on **hotpotqa**. Deltas are measured against **agentic_full** ($\dagger$), so a *negative* Δ F1 means the removed component was helping. **Bold** marks an ablation significantly *better* than the full system and † one significantly worse (paired bootstrap, 95% CI of the difference excluding zero); unmarked differences are not significant. Every row covers the same frozen 250-question evaluation slice as the main results (**agentic_full** $n = 250$, **agentic_no_planner** $n = 250$, **agentic_no_kg** $n = 250$, **agentic_no_verifier** $n = 250$, **hybrid_rerank** $n = 250$). **hybrid_rerank** is repeated as the non-agentic floor. Latency is the median per question. - across a whole row marks a configuration that has not been run. **hybrid_rerank** is retrieval-only: it emits a ranking and no answer, so the EM, F1, Δ F1 and p columns show - because there is no answer to score, not because the run is absent. Scoring the unattempted answer would report 0.000 and assert a failed attempt that never happened.

System	n	EM	F1	Δ F1 vs. full [95% CI]	p	LLM calls	Latency (s) med.
agentic_full	250	0.432 †	0.510 †	-	-	4.26	28.3
agentic_no_planner	250	0.456	0.543	+0.033 [-0.017, +0.086]	0.186	1.35	9.7
agentic_no_kg	250	0.404	0.501	-0.009 [-0.048, +0.029]	0.646	4.28	11.8
agentic_no_verifier	250	0.396 †	0.474 †	-0.037 [-0.065, -0.011]	0.008	2.06	16.5
hybrid_rerank (non-agentic floor)	250	-	-	-	-	0.00	4.9

Three ablations attribute the behaviour to its parts (Tables 4.7 and 4.8), each over the same 250 questions so every delta is paired.

Table 4.8: Ablation study on `twowiki`. Deltas are measured against `agentic_full` ($\dagger$), so a *negative* Δ F1 means the removed component was helping. **Bold** marks an ablation significantly *better* than the full system and † one significantly worse (paired bootstrap, 95% CI of the difference excluding zero); unmarked differences are not significant. Every row covers the same frozen 250-question evaluation slice as the main results (`agentic_full` $n = 250$, `agentic_no_planner` $n = 250$, `agentic_no_kg` $n = 250$, `agentic_no_verifier` $n = 250$, `hybrid_rerank` $n = 250$). `hybrid_rerank` is repeated as the non-agentic floor. Latency is the median per question. - across a whole row marks a configuration that has not been run. `hybrid_rerank` is retrieval-only: it emits a ranking and no answer, so the EM, F1, Δ F1 and p columns show - because there is no answer to score, not because the run is absent. Scoring the unattempted answer would report 0.000 and assert a failed attempt that never happened.

System	n	EM	F1	Δ F1 vs. full [95% CI]	p	LLM calls	Latency (s) med.
<code>agentic_full</code>	250	0.224 †	0.267 †	-	-	3.80	35.4
<code>agentic_no_planner</code>	250	0.324	0.388	+0.121 [+0.067, +0.173]	<0.001	1.26	4.5
<code>agentic_no_kg</code>	250	0.256	0.285	+0.018 [-0.015, +0.053]	0.252	3.87	15.8
<code>agentic_no_verifier</code>	250	0.216	0.266	-0.001 [-0.025, +0.025]	1.000	2.03	18.4
<code>hybrid_rerank</code> (<i>non-agentic floor</i>)	250	-	-	-	-	0.00	8.3

The verifier. This is the measurement the project exists to make. With the backward edge removed and everything else fixed, HotpotQA exact match falls from 0.432 to 0.396 and F1 from 0.510 to 0.474, a paired Δ F1 of -0.037 $[-0.065, -0.011]$, $p = 0.008$. The loop pays for itself on this dataset, at roughly double the model calls (4.26 against 2.06) and the median latency (28.3s against 16.5s). Then 2WikiMultihopQA does not agree. `agentic_no_verifier` scores 0.216 and 0.266 against the full system’s 0.224 and 0.267, a Δ F1 of -0.001 $[-0.025, +0.025]$, $p = 1.000$. It is not a failure to fire: the loop ran on 98 of 250 questions and a later cycle was selected on 64, and it charged the same doubled cost for a change of one thousandth of an F1 point. The honest statement is the conditional one. The backward edge helps on HotpotQA and not on 2WikiMultihopQA, and one more dataset would not make it a general result. Two readings fit: a system at F1 0.267 is failing for reasons a second attempt at the same corpus cannot repair, and 2WikiMultihopQA’s comparison templates defeat an entailment check that reads one answer sentence, since “A was born before B” is entailed by neither operand’s page alone. The ablated system being better attributed there, supporting-fact F1 0.465 against 0.421, supports the second.

The planner. This is the surprise, and the project’s most consequential negative result. `agentic_no_planner` scores above the full system on HotpotQA, exact match 0.456 and F1 0.543 against 0.432 and 0.510, Δ F1 $+0.033$ $[-0.017, +0.086]$, not significant, at 1.35 model calls and a median of 9.7s. On 2WikiMultihopQA it is decisive: 0.324 and 0.388 against 0.224 and 0.267, Δ F1 $+0.121$ $[+0.067, +0.173]$, $p < 0.001$, at 1.26 calls against 3.80 and 4.5s against 35.4s. Removing the component the brief lists first is worth twelve F1 points on the harder dataset at a third of the calls and an eighth of the wall clock, and the same row is the only configuration in the grid to beat `self_ask` significantly, Δ F1 $+0.078$ $[+0.021, +0.135]$, while grounding 0.900 of its citations. What that configuration is matters. With the Planner removed the plan is one identity node, so it issues a single query exactly as `naive_rag` does, and `naive_rag` scores 0.152 and 0.200 on the same questions. The gap between two systems that each ask one question is the rest of the agentic stack: graph evidence beside the retrieved passages, sentence-granular pooling, synthesis constrained to cite what it used, and a verification pass. The evidence handling earns its cost; the decomposition was spending it. The cost grows with chain length, which rules out the reading that decomposition merely dilutes a good ranking, since 2WikiMultihopQA’s rankings are the worse of the two. What grows with chain length is the number of sub-answers an 8B model must compose across, and each composition is a chance to lose the thread.

The graph. A null was always plausible, since the induced graph is a navigation aid rather than a knowledge base, and a null is what returns. `agentic_no_kg` scores 0.404 and 0.501 on HotpotQA, Δ F1 -0.009 $[-0.048, +0.029]$, and 0.256 and 0.285 on 2WikiMultihopQA, Δ F1 $+0.018$ $[-0.015, +0.053]$, with retrieval statistically indistinguishable from the full system’s. What the graph changes is the bill: tool calls fall from 14.83 to 8.15 per question and median latency from 28.3s to 11.8s on HotpotQA, mirrored on 2WikiMultihopQA. The KG Navigator accounts for roughly half the system’s tool calls and more than half its latency and returns nothing either evaluation can detect. Unlike the planner this is a null rather than a harm: the graph is paid for and ignored.

4.6 Error Analysis

Table 4.9: Failure profile on **hotpotqa**. Labels are assigned post hoc from the trace and the gold supporting facts, first match winning, from the system never having a chance to having everything and still getting it wrong. **verifier_false_accept** and **verifier_false_reject** are counted again in the totals, independently of the cascade, since they are what make the feedback loop falsifiable. A configuration with no run has no column. **bm25_only**, **dense_only** and **hybrid_rerank** emit no answer at all, so their *Correct* is structurally 0 and every question is counted as an error; the informative rows for them are the retrieval labels, not the accuracy.

Label	bm25_only	dense_only	hybrid_rerank	naive_rag	self_ask	agentic_full	agentic_no_planner	agentic_no_kg	agentic_no_verifier
<i>Failure labels: count (share of all questions)</i>									
parse_failure	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	1 (0.004)	0 (0.000)
budget_exhausted	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)
retrieval_miss	4 (0.016)	5 (0.020)	0 (0.000)	0 (0.000)	0 (0.000)	3 (0.012)	1 (0.004)	3 (0.012)	3 (0.012)
decomposition_error	105 (0.420)	82 (0.328)	62 (0.248)	44 (0.176)	31 (0.124)	16 (0.064)	46 (0.184)	15 (0.060)	3 (0.012)
bridge_link_failure	104 (0.416)	115 (0.460)	138 (0.552)	55 (0.220)	69 (0.276)	17 (0.068)	63 (0.252)	15 (0.060)	20 (0.080)
synthesis_error	17 (0.068)	42 (0.168)	33 (0.132)	19 (0.076)	16 (0.064)	57 (0.228)	23 (0.092)	60 (0.240)	61 (0.244)
verifier_false_accept	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	33 (0.132)	0 (0.000)	36 (0.144)	0 (0.000)
verifier_false_reject	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	1 (0.004)	0 (0.000)
<i>Totals</i>									
Questions scored	250	250	250	250	250	250	250	250	250
Correct	0	0	0	124	126	108	114	101	99
Accuracy	0.000	0.000	0.000	0.496	0.504	0.432	0.456	0.404	0.396
Errors	250	250	250	126	124	142	136	149	151
Errors with no rule firing	20	6	17	8	8	16	3	18	64
Verifier false accepts	0	0	0	0	0	92	74	99	0
Verifier false rejects	0	0	0	0	0	2	0	5	0
Recoverable (some cycle was right)	0	0	0	0	0	2	0	5	0

Table 4.10: Failure profile on **twowiki**. Labels are assigned post hoc from the trace and the gold supporting facts, first match winning, from the system never having a chance to having everything and still getting it wrong. **verifier_false_accept** and **verifier_false_reject** are counted again in the totals, independently of the cascade, since they are what make the feedback loop falsifiable. A configuration with no run has no column. **bm25_only**, **dense_only** and **hybrid_rerank** emit no answer at all, so their *Correct* is structurally 0 and every question is counted as an error; the informative rows for them are the retrieval labels, not the accuracy.

Label	bm25_only	dense_only	hybrid_rerank	naive_rag	self_ask	agentic_full	agentic_no_planner	agentic_no_kg	agentic_no_verifier
<i>Failure labels: count (share of all questions)</i>									
parse_failure	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)
budget_exhausted	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)
retrieval_miss	4 (0.016)	0 (0.000)	1 (0.004)	1 (0.004)	1 (0.004)	1 (0.004)	1 (0.004)	1 (0.004)	0 (0.000)
decomposition_error	147 (0.588)	144 (0.576)	138 (0.552)	124 (0.496)	48 (0.192)	22 (0.088)	108 (0.432)	25 (0.100)	0 (0.000)
bridge_link_failure	45 (0.180)	49 (0.196)	50 (0.200)	42 (0.168)	81 (0.324)	7 (0.028)	36 (0.144)	8 (0.032)	9 (0.036)
synthesis_error	36 (0.144)	45 (0.180)	46 (0.184)	34 (0.136)	19 (0.076)	59 (0.236)	22 (0.088)	52 (0.208)	59 (0.236)
verifier_false_accept	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	82 (0.328)	1 (0.004)	79 (0.316)	0 (0.000)
verifier_false_reject	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)	0 (0.000)
<i>Totals</i>									
Questions scored	250	250	250	250	250	250	250	250	250
Correct	0	0	0	38	66	56	81	64	54
Accuracy	0.000	0.000	0.000	0.152	0.264	0.224	0.324	0.256	0.216
Errors	250	250	250	212	184	194	169	186	196
Errors with no rule firing	18	12	15	11	35	23	1	21	128
Verifier false accepts	0	0	0	0	0	154	102	144	0
Verifier false rejects	0	0	0	0	0	5	0	5	0
Recoverable (some cycle was right)	0	0	0	0	0	5	0	5	0

Failures are categorised from the traces against the gold supporting facts, first match winning (Tables 4.9 and 4.10). Neither dataset records a single parse failure or budget exhaustion in any configuration, which retires two concerns the design spent effort on. Against **hybrid_rerank**, **agentic_full** reduces the share of questions labelled decomposition error from 0.248 to 0.064 on HotpotQA and from 0.552 to 0.088 on 2WikiMultihopQA, and bridge-link failure from 0.552 to 0.068 and 0.200 to 0.028. Those are the two classes Chapter 1 argued a single-shot ranker cannot address, and they have very largely gone. What replaces them is synthesis error, at 0.228 and 0.236, the dominant agentic failure mode and more than three times the rate for **self_ask**. The system finds the right evidence and composes the wrong string from it.

The Verifier’s behaviour is the other half of that picture. The traces record 92 false accepts on HotpotQA and 154 on 2WikiMultihopQA against 2 and 5 false rejects. The loop is far too permissive, and the confidence signal is why: a post-hoc sweep over the calibration slice finds an area under the ROC curve of 0.633 and an expected calibration error of 0.332, a usable ranking cut-point and an unusable probability. The cleanest single failure is HotpotQA question **5a7af32e55429931da12c99c**: the system answered New York City where the gold is Brooklyn, New York, with an answer sentence copied from an unrelated evidence sentence, and the Verifier scored entailment above 0.99 because a

copied sentence is trivially entailed by its source. The architecture entails the answer sentence and never the answer string, and the two can come apart.

4.7 Discussion

The grid does not support the claim this project would have preferred to make. Agency did not buy accuracy. A published decomposition method with no verification answers more questions correctly than the four-agent system at a third of the calls, and the fused ranking the agentic system produces is significantly worse than a single reranked query. Agency did buy grounded, auditable answers, and the failure profile says where the effort belongs: not in more planning, more traversal or more retrieval, all of which already deliver the evidence, but in synthesis, where the right sentences become the wrong span. A practitioner should take the following. For a ranked list, use **hybrid_rerank**. For a short answer where cost matters, use **self_ask**. The four-agent system earns its cost only where the answer must be attributable. And the one-node configuration that deletes the Planner is the system to start from, because it gives up nothing.

4.8 Improving the System, and What It Cost

That last sentence was the starting point for a bounded improvement loop: five hypotheses fixed in advance, each measured on the same frozen 250 questions and kept or dropped on a paired interval, under stop conditions stated before any ran. One local model, no paid API, and the nine evaluated configurations were not to change: every modification ships behind a flag defaulting to the evaluated behaviour, and the four evaluated agentic configurations are checked to resolve to identical settings before and after, so **agentic_full** remains the system this chapter describes. One hypothesis survived.

4.8.1 The Kept Change

qwen3:8b is a hybrid reasoning model whose reasoning mode is suppressed throughout the grid, for the reason Section 3.4 gave. The kept configuration, **agentic_v2**, re-enables it for exactly one agent. The synthesiser reasons; the Planner, Retrieval Agent, KG Navigator and Verifier do not, through a per-agent routing list added beside the global flag. The configuration also raises the completion budget from 1024 to 5120 tokens, and that is part of the change: at 1024 the reasoning block consumes the whole budget and the synthesiser returns nothing. It builds on the one-node plan, since the ablation above showed decomposition subtracting from the answer.

Table 4.11: The improvement loop’s kept configuration against the reported system, its own base, and the strongest answer baseline. Paired on question id over the same frozen 250-question slice. Five of the six intervals exclude zero; the exception is **self_ask** on HotpotQA, where the kept configuration ties a single-model baseline that has no agents in it.

System	<i>n</i>	EM	F1	Δ EM vs. kept [95% CI]	<i>p</i>	gained / lost
<i>HotpotQA</i>						
agentic_v2 (kept)	250	0.524	0.618	–	–	–
agentic_full	250	0.432	0.510	+0.092 [+0.036, +0.144]	<0.001	36 / 13
agentic_no_planner	250	0.456	0.543	+0.068 [+0.024, +0.112]	0.002	25 / 8
self_ask	250	0.504	0.603	+0.020 [−0.032, +0.072]	0.499	25 / 20
<i>2WikiMultihopQA</i>						
agentic_v2 (kept)	250	0.456	0.511	–	–	–
agentic_full	250	0.224	0.267	+0.232 [+0.168, +0.296]	<0.001	67 / 9
agentic_no_planner	250	0.324	0.388	+0.132 [+0.080, +0.188]	<0.001	42 / 9
self_ask	250	0.264	0.310	+0.192 [+0.128, +0.256]	<0.001	64 / 16

Table 4.11 gives the outcome, paired over the same 250 questions with 2000 bootstrap resamples. Against the system this chapter describes, exact match rises from 0.432 to 0.524 on HotpotQA and from 0.224 to 0.456 on 2WikiMultihopQA, both intervals well clear of zero. Against its own base, the

one-node plan without reasoning, the gain is +0.068 and +0.132. The comparison that matters most for the architectural claim is the one that does not clear zero. On HotpotQA the kept configuration gains +0.020 over `self_ask`, interval $[-0.032, +0.072]$, $p = 0.499$: it ties a single-model baseline that contains no agents at all. On 2WikiMultihopQA the same comparison gives +0.192. The architecture earns its place on the dataset built to require multiple hops and does not earn it on the one that does not. That is the conclusion the ablations already reached, reached again at a higher score, and the improvement does not overturn it.

Table 4.12: Cost per question. The kept configuration answers with roughly a third of the model calls and a faster median, and pays for it in the tail: a synthesis call that reasons may run to the full 5,120-token completion budget.

Dataset	System	Median (s)	p90 (s)	LLM calls
HotpotQA	agentic_v2 (kept)	18.7	115.6	1.37
	<code>agentic_full</code>	28.3	55.4	4.26
2WikiMultihopQA	agentic_v2 (kept)	32.2	144.7	1.21
	<code>agentic_full</code>	35.4	69.2	3.80

Table 4.12 gives the cost. The kept configuration answers with a third of the model calls of `agentic_full` and a faster median, and pays in the tail: a synthesis call that reasons may run to the full budget, and the p90 roughly doubles. For a system whose latency was already the binding practical constraint, that trade is stated in both directions.

4.8.2 What the Change Costs That the Scores Do Not Show

Table 4.13: What reasoning on the synthesis call costs. The model spends its entire generation budget inside the thinking channel and never opens the content channel (`done_reason_length` at the `num_predict` cap), and the extractive fallback answers instead. Every one of these losses is already inside the exact-match figures of Table 4.11.

Condition	n	Synthesis calls	Questions losing one	Questions retrying
HotpotQA, reasoning off	250	250	0 (0.0%)	0 (0.0%)
2WikiMultihopQA, reasoning off	250	250	0 (0.0%)	0 (0.0%)
HotpotQA, reasoning on	250	251	2 (0.8%)	26 (10.4%)
2WikiMultihopQA, reasoning on	250	251	8 (3.2%)	49 (19.6%)

Enabling reasoning introduces a failure mode that suppressing it does not have, and Table 4.13 counts it. Across 500 questions with reasoning off, no synthesis call failed. Across the same 500 with it on, ten failed outright and seventy-five more needed a retry the other condition never needed. The mechanism was confirmed by instrumenting the client: the call ends with Ollama’s `done_reason` equal to `length` at exactly the configured cap. The model spends its entire budget inside the thinking channel and never opens the content channel, and because Ollama returns reasoning out of band the visible reply is genuinely empty rather than cut mid-sentence. The extractive fallback rule then supplies a confident wrong answer.

Two things follow. Those ten losses are already inside the reported figures, each scoring zero in the runs that produced 0.524 and 0.456, so the gains are net of them. And the obvious repair does not work: re-issuing each failed call with reasoning disabled recovers the content channel ten times out of ten and produces zero correct answers. The retries are a latency cost, not an accuracy one. The failure was invisible for two rounds of the loop because the trace declared a `truncated` field that no code ever assigned, and the loop concluded twice, in writing, that the failures were not the cap. Populating the field immediately exposed a second defect, in which the harness built its model client from the shipped configuration rather than the run’s. Both are fixed and tested, and both are the kind of defect that only instrumentation finds.

4.8.3 What Was Dropped

Four hypotheses were dropped, two of them without spending any GPU time, which is the loop’s most useful property.

A larger local model. `qwen3:14b` runs at 11.1 tokens per second on this hardware against the 24.5 the evaluation budget requires, with 58% of the model resident on the GPU, and freeing key-value cache bought no additional GPU layers because the runtime pins the resident fraction. The fallback, `gemma3:12b`, has no reasoning mode and cannot reproduce the one change that worked.

A wider evidence budget. The plan held that the synthesiser’s 20-item evidence cap was binding, citing a gain from 152 to 155 complete gold pools at 30. An offline replay showed the premise was false for the system being improved: the one-node plan returns a median of ten sentences, only three questions in 250 exceed twenty items, and the cap moves complete pools from 142 to 143. The 152 figure described the five-sub-query pool of `agentic_full`, carried across from a different system. The same replay closed a reserved graph quota: knowledge-graph rows score between 0.2255 and 1.0000 in the evidence ranking while passage sentences fall in 0.0333 to 0.1000, so every triple outranks every passage and a wider budget fills the new slots with triples above the sentences carrying the answer.

Self-consistency voting. Sampling the synthesiser three times was costed at 6.5 GPU hours for HotpotQA, and the failure mode it targets was counted first. Of 119 wrong answers, nineteen have the shape voting suppresses, the gold string present in the model’s own answer sentence but a different span in the answer field. Fifteen are surface-form artifacts, `ten` against `10` or `Los Alamos Laboratory` against `Los Alamos`, with no variance for voting to remove. Four are genuine, a ceiling of +0.016 exact match that the run-to-run movement of Section 4.8.4 cannot resolve at this sample size. It was dropped on the count rather than the run.

The loop ended on its stated condition, two consecutive drops after the performance floor had been reached, with the backlog fully decided.

4.8.4 Seed Pinning Is Not Reproducibility

Every measurement in this report is made at temperature zero with a fixed seed. That is not sufficient for determinism on this stack, and the loop established why. Issuing an identical prompt repeatedly, in one process, with the same seed and the same model, produces different completions. The prompt token count reported for the same first-attempt prompt varied across repetitions, 1218, 2977 and 3671, because Ollama’s key-value cache reuses the shared prefix of a preceding request. A different prefix gives a different batch shape, a different batch shape gives different floating-point arithmetic, and the sampling that follows diverges even when it is nominally greedy.

This supplies the mechanism for an observation Chapter 5 reports: re-running byte-identical code over the same questions changes a material fraction of answers. It is not a defect in the harness and no seed discipline removes it. Its consequence applies to every comparison in this report. At this sample size an effect of a point or two of exact match cannot be distinguished from the harness itself. Differences well above that, the gains of Table 4.11 and the planner ablation, survive it. Differences below it, including several reported here as nulls, are correctly described as unresolved and would remain so under any number of repetitions of the same design.

Chapter 5

Conclusions and Future Work

5.1 Summary of Contributions

This project set out to test a narrow claim: that adding one backward edge to a retrieval-augmented pipeline changes what the system can do, and that the change can be measured rather than described. What was built is a four-agent system coordinated by an explicit state machine, covering all five tasks of the brief’s Table 1 and running on one consumer GPU with no paid API. The Planner emits a validated graph of sub-queries. The Retrieval Agent routes each to BM25, dense or hybrid search, fuses formulations with reciprocal rank fusion [1] and reranks only when the margin is narrow. The KG Navigator links entities by alias and runs bidirectional search over a graph induced from corpus text. The Verifier resolves citations, scores entailment, and below its threshold returns a directive to the Planner instead of an answer.

Three properties were established during construction rather than by the benchmark. Schema-constrained planning with a local 8B model is reliable on structure, 8 of 8 decompositions correctly shaped with no retries, and unreliable on self-description, 7 of 8 strategy labels wrong, which is why strategy is derived from the graph and the model’s label discarded. Agency is affordable here only because most of it is deterministic. And, the finding this report did not expect to make, the model’s reasoning mode, suppressed everywhere for the grid, is worth more than any component of the architecture when it is switched on for one agent.

5.2 Empirical Findings

The evaluation answers four questions, each of which had a defensible negative answer, and the protocol was fixed before any was resolved.

Does decomposition beat single-shot retrieval? No, and it is a significant no on one of two datasets. Against `self_ask`, the strongest baseline that answers, `agentic_full` is -0.093 F1 $[-0.145, -0.039]$ on HotpotQA and -0.043 $[-0.100, +0.014]$ on 2WikiMultihopQA. On evidence the answer is different in kind: scored the same way for every row, on the recall of the whole pool, the agentic system leads every baseline significantly, and its citations resolve to real sentences at 0.960 and 0.976 where no baseline’s resolve at all. Decomposition improves which evidence is cited, and how checkably, and degrades where in a fused list it appears.

Does the backward edge pay for itself? Conditionally. Removing it costs 0.037 F1 on HotpotQA at $p = 0.008$ and 0.001 on 2WikiMultihopQA at $p = 1.000$, while the loop fired on nearly half the questions of both and doubled the cost of both. The counterweight is small, 2 and 5 false rejects, so the answer selection is doing its job; the failure is the other way, 92 and 154 false accepts, and it is a property of a confidence signal with an AUC of 0.633 rather than of where the line was drawn. The edge helps on one dataset and not the other, and this report states it that way.

What does agency cost, and where does it fail? About three times the model calls of the strongest baseline and rather more in wall clock, for a result that is worse on the answer string. It fails at composition, having stopped failing at retrieval: decomposition and bridge-link errors largely

disappear against the ranker, and synthesis error at 0.228 and 0.236 replaces them. The system finds the right sentences and extracts the wrong span.

What actually improved it? Not planning, not traversal, not a wider evidence budget, not a larger model, and not voting. Deleting the Planner was worth twelve F1 points on 2WikiMultihopQA at a third of the cost. Letting the synthesiser reason, on the one-node plan, then took exact match from 0.432 to 0.524 on HotpotQA and from 0.224 to 0.456 on 2WikiMultihopQA, ties the strongest baseline on the first and beats it by 0.192 on the second, with one change and one mechanism. Every other hypothesis was dropped on its own evidence, two of them before any GPU time was spent.

5.3 Limitations

Sample size. Results come from 250-question stratified samples. The paired intervals are the right test and they decide significance, but nothing here should be read as a precise effect size.

The runs are not exactly reproducible. Every interval in this report is a bootstrap over questions, which answers what a different 250 would have shown and is silent about what the same system does on the same questions run again. With temperature 0 and a fixed seed the intended answer was nothing.

Table 5.1: Run-to-run drift: the same configuration, the same questions, the same seed and a decoding temperature of 0.0, executed twice. *Flipped* counts questions whose exact match changed between the two executions, and the column beside it how many of those also drew a different evidence pool, locating the divergence in retrieval rather than in generation. One cause is the `PYTHONHASHSEED` column: the evidence pool is assembled from sets, whose iteration order follows Python’s per-process string hashing unless the variable is set. This is the report’s own error bar on execution, separate from every bootstrap interval elsewhere, which resample questions and hold the execution fixed. An effect smaller than this drift is not established by a single run.

Configuration	Dataset	n	EM run 1	EM run 2	Δ EM	Δ F1	Flipped	PYTHONHASHSEED
<code>agentic_no_verifier</code>	<code>hotpotqa</code>	250	0.396	0.384	-0.012	-0.014	15 (13 via evidence)	<code>unset vs 0</code>

Table 5.1 shows it is not nothing. `agentic_no_verifier`, a configuration with no feedback loop to blame, was re-run on the identical HotpotQA slice: exact match moved by 0.012, F1 by 0.014, and 15 of 250 answers changed. The runs’ own metadata records one cause, an unpinned Python hash seed that let evidence pools assemble from sets in varying order, and the harness now pins it. Section 4.8.4 records the deeper one, which pinning does not remove: on a local inference server with prefix caching, identical prompts at temperature zero produce different completions, and on the calibration slice between 6 and 14 percent of answers change between executions of byte-identical code. The consequence is that an effect of a point or two is not established by a single execution. The planner ablation and the kept configuration’s gains are comfortably clear of that floor; the graph’s two nulls and the verifier’s 2WikiMultihopQA null are inside it and remain nulls. Every interval elsewhere is narrower than the true uncertainty, because it holds the execution fixed.

Model scale and domain. Every result is conditional on a 4-bit 8B model, and the improvement loop measured that a 14B model does not fit this hardware at an acceptable throughput. Both benchmarks are Wikipedia: clean, segmented and densely linked, which the entity graph exploits. Nothing here is evidence of transfer to legal, biomedical or enterprise collections.

The Verifier checks the sentence, not the answer. The entailment hypothesis is the synthesiser’s answer sentence; the answer string the metrics score is never entailed against anything, and the trace record in Section 4.6 shows the two coming apart at a confidence of 0.82. The correction, entailing a hypothesis built from the question and the answer string, is straightforward and was not in place for any run reported here.

An uncalibrated threshold. The threshold of 0.55 was set a priori. The post-hoc sweep over the calibration slice found the confidence blend to be a usable ranking cut-point and an unusable probability, with an optimal threshold whose bootstrap interval spans 0.45 to 0.64 and does not identify an operating point. A genuine sweep needs one full run per candidate, since the threshold changes what re-plans and therefore what is answered.

Re-planning overwrites state, and the budget is cooperative. Sub-query identifiers restart at `q1` in every revision, so a re-plan overwrites the previous cycle’s results and the trace cannot fully reconstruct what the synthesiser saw in cycle zero. The 300 s budget is tested between calls and cannot pre-empt one, which is how a question ran for 18.3 hours inside a suspended call. Both are recorded defects, not applied to any run here.

Silent degradation. If the entailment model cannot load, the Verifier falls back to token containment, which scored 1.000 where the model scores 0.0008; every verdict becomes accept and the loop never fires. A preflight check now refuses to start without the encoder. The general lesson outweighs the bug: the fallbacks that keep a long run alive are exactly the ones that quietly change what is measured, and every degradation path needs a gate or a trace flag. The unassigned `truncated` field of Section 4.8 is the same lesson from the other side.

5.4 Future Work

Synthesis, not planning. The evidence points one way. The next unit of effort belongs in answer extraction and in a verifier that can distinguish a well-cited answer from a correct one: atomic claim decomposition, a hypothesis that includes the answer string, and cross-source agreement, so the directive tells the Planner what to look for rather than only that something is missing.

Learned rather than heuristic decisions. The routing table, the rerank gate and the confidence blend are hand-specified functions whose inputs are logged on every question. The cheap first step is offline: the trace records the Planner’s hint and the lexical features on every sub-query, so the agreement between the hint that was honoured and the rule that would have fired can be computed without a run, and it would say whether six of seven rules are redundant or merely unreached.

Evaluation on a caching inference server. The reproducibility finding generalises beyond this project. Any evaluation of this size on a local server with prefix caching carries a noise floor that seed discipline does not remove, and the honest protocol is several executions per configuration with the spread reported, not one execution with a seed. That is a half day of compute here and it is the first thing a continuation should do.

Domain transfer. Biomedical corpora have curated ontologies, so the KG Navigator could traverse an authored graph and isolate how much of a graph’s value comes from structure rather than induction. Legal corpora invert the challenge, with explicit citation networks but long, cross-referential passages that would test alias linking to destruction.

5.5 Toward a Thesis Extension

Two of the thesis directions in the brief follow from this system. The self-reflective fact-checking track is the closer fit: the Verifier already scores confidence and triggers further retrieval when it is low, and what it lacks is multi-source cross-verification, provenance annotation and calibrated uncertainty reported to the user. A multimodal setting needs a cross-modal entailment model in place of the NLI stage and leaves the orchestration intact. The query-decomposition track is the other: a spatio-temporal version decomposes over archives indexed by time and location with the same validated graph and feedback loop, and the cost model becomes the central design problem once one tool invocation is a satellite-imagery request rather than a BM25 query. Both would be the natural place for the real-time external tools the brief describes and this system deliberately omits.

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